

- 3 (a) By considering the binomial expansion of $(z+z^{-1})^4$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that $\cos^4 \theta = \frac{1}{8}(\cos 4\theta + 4 \cos 2\theta + 3)$. [5]

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- (b) Use the substitution $x = \sin \theta$ to find the exact value of $\int_0^{\frac{1}{2}} (1-x^2)^{\frac{3}{2}} dx$. [3]

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